

Shekhar Sen

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SUMMARY

Results-driven Software Engineer with 3+ years of experience in developing AI-powered applications, automation solutions, and scalable web interfaces using Python, SQL, React.js, and REST APIs. Skilled in building customer-centric solutions, optimizing system performance, and troubleshooting complex technical issues. Strong analytical and problem-solving abilities with a focus on delivering reliable, production-ready software solutions.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

Frameworks: React, Node.js, Flask, Material-UI, Fast API **Frontend**

Technologies: React.js, HTML5, CSS3, Material UI

Backend Technologies & Frameworks: Node.js, Flask, FastAPI, REST APIs, JWT Authentication

Developer Tools & Platforms: Git, Docker, AWS, Google Cloud Platform (GCP), CI/CD (GitHub Actions, Jenkins, GitLab CI), Kubernetes, Visual Studio Code, Visual Studio, PyCharm

Databases: PostgreSQL, Microsoft SQL Server, MongoDB, Pinecone, ChromaDB, FAISS

Data Science & Machine Learning Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Generative AI & LLM Technologies: Retrieval-Augmented Generation (RAG), Prompt Engineering, LangGraph, LangChain, CrewAI, Hugging Face Transformers, Embeddings, Natural Language Processing (NLP)

Deep Learning: TensorFlow, Keras, CNN, PyTorch

EXPERIENCE

Senior Software Engineer

12/2022 - Present

Capgemini Technology Services India LTD.

Bengaluru, KA

- Engineered an AI-powered conversational chatbot using **Python, NLP, Machine Learning, LangChain, LangGraph, and Retrieval-Augmented Generation (RAG)**, leveraging client-specific datasets to automate approximately 65–75% of repetitive customer queries and reduce average response time from ~8 minutes to under 1 minute.
- Trained, evaluated, and optimized NLP/ML intent classification models using 15K+ historical conversation records, applying **Python, Pandas, NumPy, and Scikit-learn** to achieve approximately 80–88% intent recognition accuracy and enable context-aware conversational interactions.
- Developed a responsive and reusable **React.js frontend using JavaScript/TypeScript, HTML5, CSS3, and Material-UI**, improving customer engagement metrics by approximately 30% and reducing user drop-off during chatbot interactions.
- Designed and developed Python-based backend microservices using **FastAPI/Flask**, exposing REST APIs to integrate frontend components with AI/NLP services and maintaining sub-3-second response latency for real-time chatbot interactions.
- Implemented secure and scalable backend communication using **REST APIs and JWT authentication**, enabling reliable integration between the React.js frontend, backend microservices, NLP/ML pipelines, and data services.
- Architected a modular, scalable, and extensible microservices-based system using **Python, FastAPI/Flask, React.js, REST APIs, Docker, and Kubernetes**, reducing feature deployment effort by approximately 40% and simplifying future enhancements.
- Containerized application components using **Docker and deployed the chatbot platform to AWS Elastic Kubernetes Service (EKS) using Kubernetes**, enabling scalable and reliable cloud-based application deployment.
- Automated CI/CD pipelines using GitHub Actions to build, test, containerize, and deploy application code to AWS EKS, improving deployment consistency and reducing manual release effort.
- Integrated application data and AI retrieval components with **PostgreSQL/MongoDB and vector databases**, supporting structured data management and semantic search for context-aware chatbot responses.
- Executed comprehensive functional, integration, API, conversational, and edge-case testing across 150+ chatbot workflows, reducing production issues by approximately 50% and improving overall system reliability.
- Used Git-based version control and CI/CD practices to manage source code, automate validation and deployment workflows, and support collaborative development across frontend, backend, ML, and infrastructure components.

EDUCATION

Sikkim Manipal Institute of Technology

Bachelor of Technology, Electronics and Communication Engineering

Majitar, Sikkim, India

July 2018 – Oct. 2022

CERTIFICATIONS

AWS Certified Cloud Practitioner